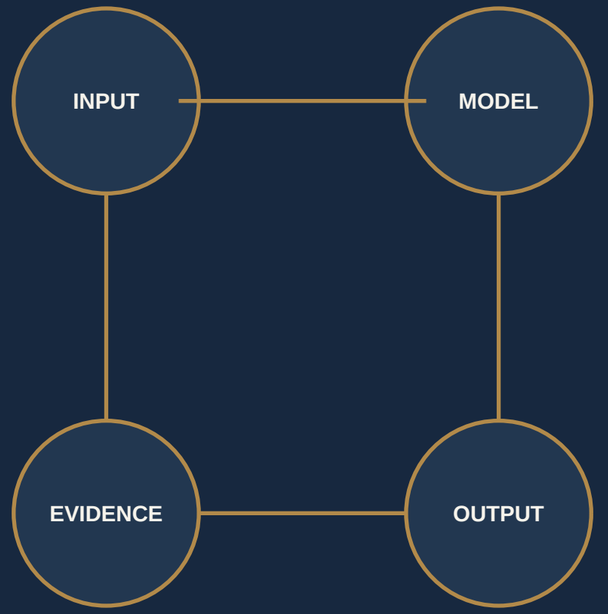
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Neural Networks

AI.111 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define neurons.
- Apply key methods including perceptrons.
- Build or analyze a practical system involving activation functions.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Neurons

02

METHODS

Perceptrons

03

SYSTEMS

Activation functions

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Neurons
Neurons

02 · Perceptrons
Perceptrons

03 · Activation functions
Activation functions

04 · Forward propagation
Forward propagation

05 · Backpropagation
Backpropagation

06 · Gradient descent
Gradient descent

07 · Multilayer networks
Multilayer networks

08 · Optimization and neural-networ
Optimization and neural-network architectures

Core ideas, made explicit

AI.111 • Distinguish definitions, assumptions and evidence

- Neurons
- Perceptrons
- Activation functions
- Forward propagation
- Backpropagation

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.111 • Distinguish definitions, assumptions and evidence

- Gradient descent
- Multilayer networks
- Optimization and neural-network architectures
- Animated neural-network diagrams
- Mandatory practicals: Build neural networks from scratch and using a modern framework

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM

Neurons

INPUTS

Perceptrons

METHOD

Activation functions

OUTPUT

Forward propagation

EVALUATION

Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving neurons.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using neurons.
- Implement or simulate a baseline using perceptrons.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.